Computer Aided Graphics

THE SOLAR SYSTEM

~GUI in Matlab~

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I. INTRODUCTION

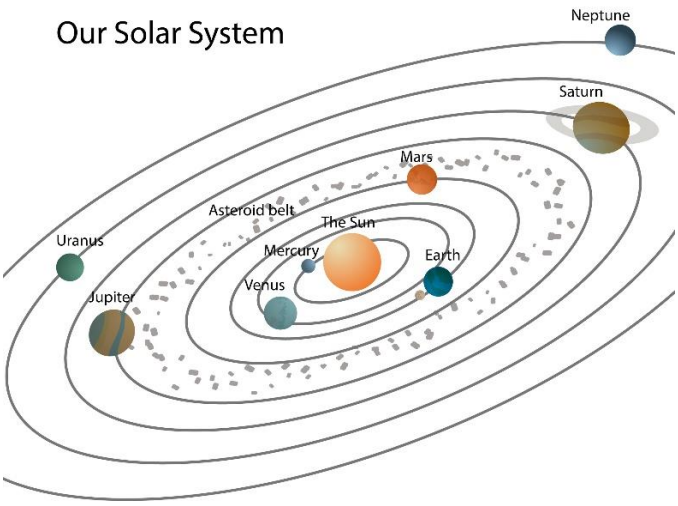
- Standing for **MAT**rix **LAB**oratory, Matlab was invented in the late 1970s by the chairman of the Computer Science department of the University of New Mexico, **Cleve Moler.** In those times, C and Fortan were widely used, but scientists and engineers were in need of a more powerful and versatile environment provided for calculus with matrices.

- Today, **Matlab** is a valuable tool, being widely used in many industries, finances and for engineering research. Nowadays it is very easy to use, very error tolerant, pictural and offers many suggestions when typing or creating functions or programs.

DID YOU KNOW THAT???

- The toolboxes **Matlab** has containing packages of built-in functions and classes, provide great help for topics like data acquisition, modelling and simulation, data and signal analysis, image processing, deep learning, statistics etc.

II. My Project: The Solar System



- The **Solar System** is the gravitationally bound system of the Sun and the objects that orbit it. It formed about 4.6 billion years ago when a dense region of a molecular cloud collapsed, forming the **Sun** and a protoplanetary disc.

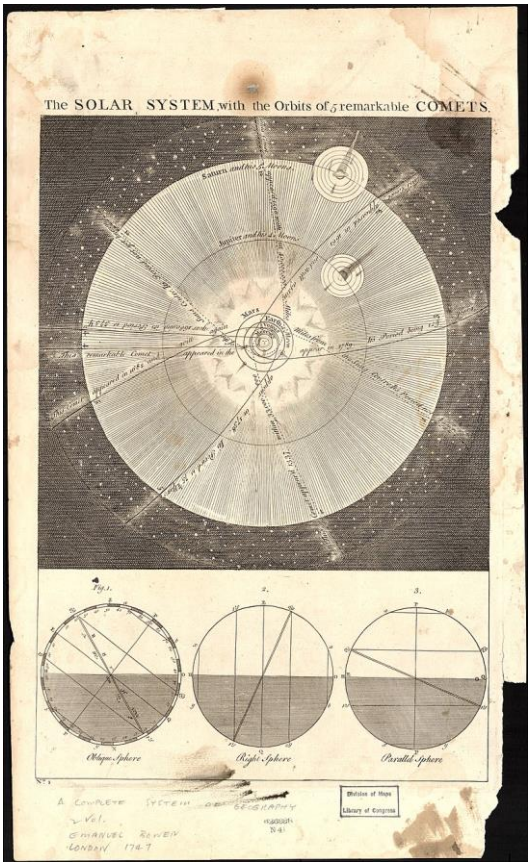
- The largest objects that orbit the Sun are the eight planets. In order from the Sun, they are four terrestrial planets (***Mercury***, ***Venus***, ***Earth*** **and** ***Mars***); two gas giants (***Jupiter*** **and** ***Saturn***); and two ice giants (***Uranus*** **and** ***Neptune***). All terrestrial planets have solid surfaces.

Definition:

How was it discovered?

- Initially, ancient and medieval astronomers observed visible objects like the Sun, Moon, planets, comets, and atmospheric phenomena. Early tools and geometric observations paved the way for the **telescope's** invention, enabling the discovery of planetary moons, rings, asteroids, and new planets. This era revealed Earth as a planet and stars as other suns, distinguishing the Solar System as a separate entity.

- For centuries, the **Solar System** was perceived as the entire universe until the mid-17th century, when the distinction between it and outer space became clear. Advances in spectroscopy and various branches of astronomy (e.g., radio, infrared, X-ray) enriched knowledge about celestial compositions. Robotic probes, the **Apollo Moon landings**, and space telescopes furthered planetary science, offering insights into planetary atmospheres, geology, and electromagnetic properties.



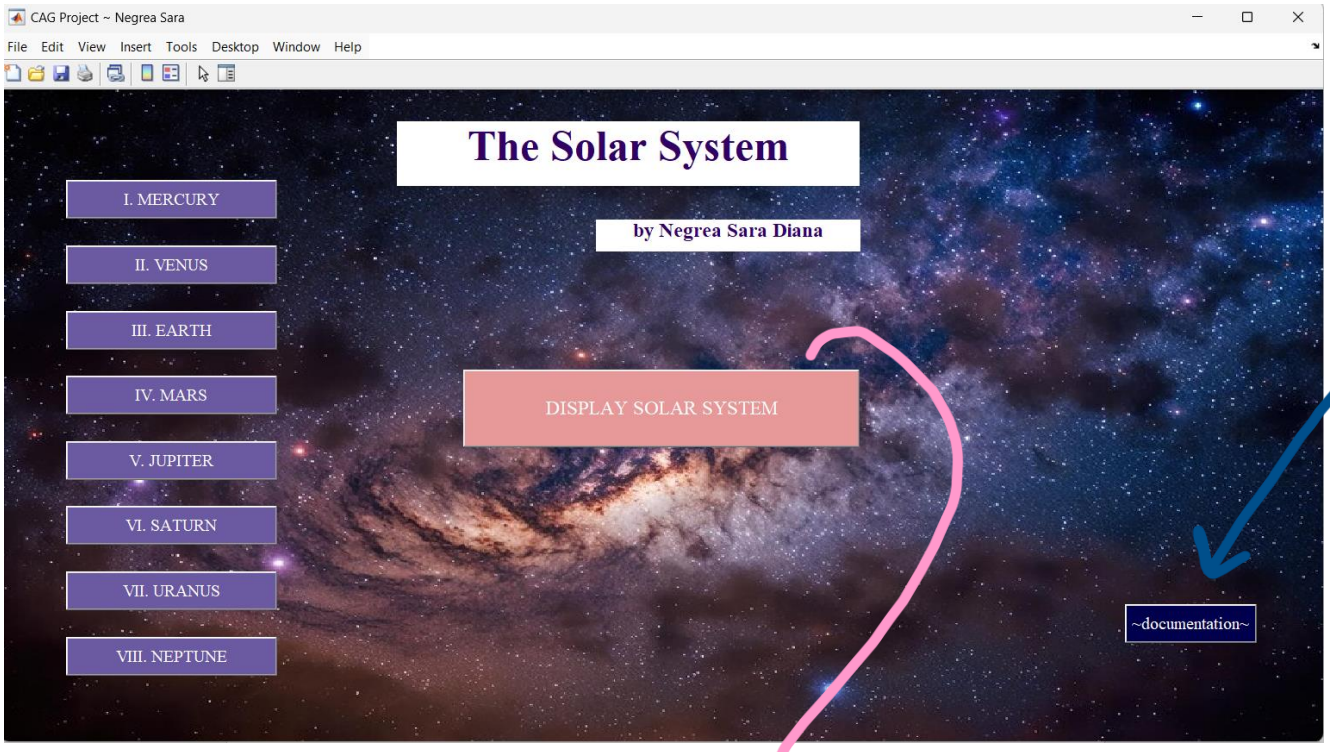
III. Theoretical presentation

- **CONCEPT:**

-the project includes a graphic interface that will show in 3D the real size of each planet of the Solar System separately, displaying its radius and its surface, followed by a real scaled graphic will all 8 planets.

- This **pushbutton** will open a word document (this one) that will contain all the details about the GUI

The main Menu

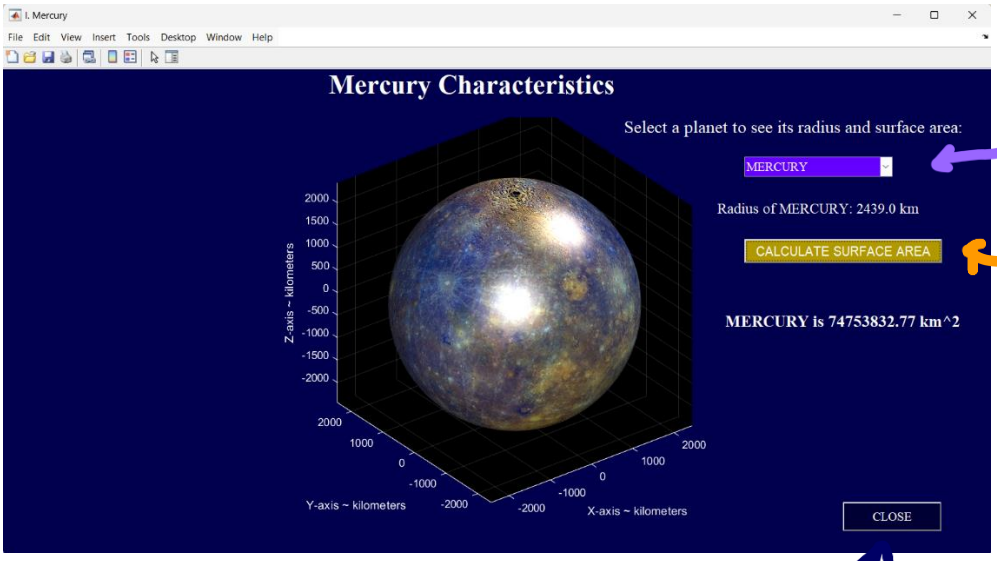


- Each of these **pushbuttons** will open a separate window for each planet

- This pushbutton will open a real-life scaled graphic with all planets

Computer Aided Graphics

○ The Planets Characteristics

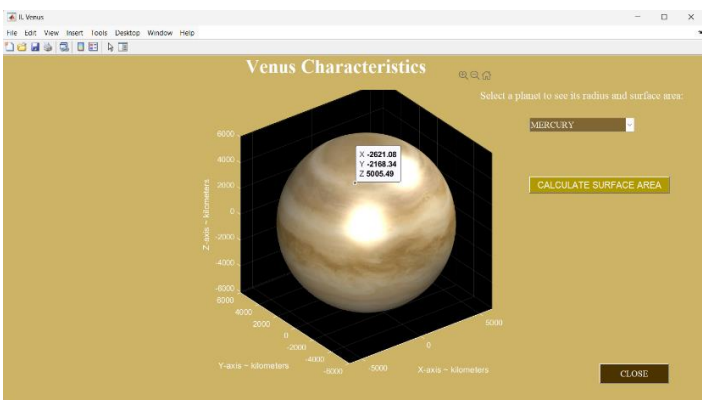


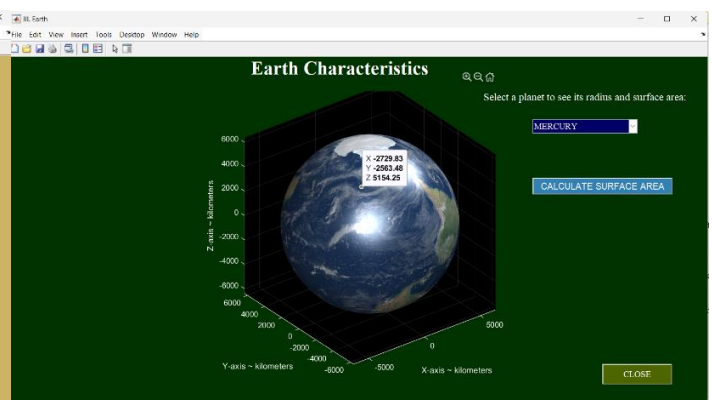
-The **popup menu** is holding the radius for each planet (+the name of it)

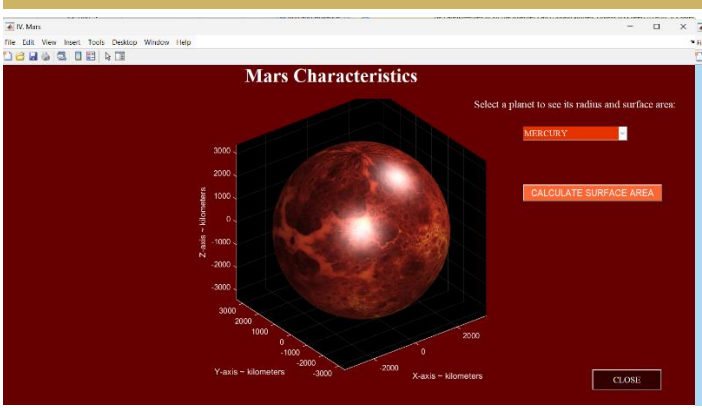
-The '**CALCULATE SURFACE AREA**' button will calculate, using the previous selected radius, the surface of the entire planet with the FORMULA : $\mathbf{A = 4\pi r^2}$ and it will display it

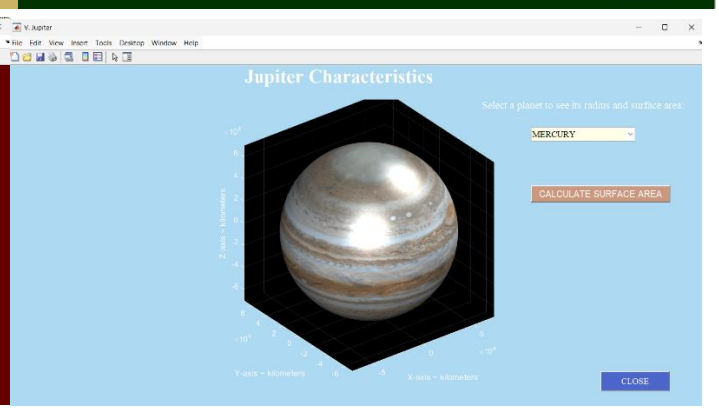
-The **close** button returns to the main

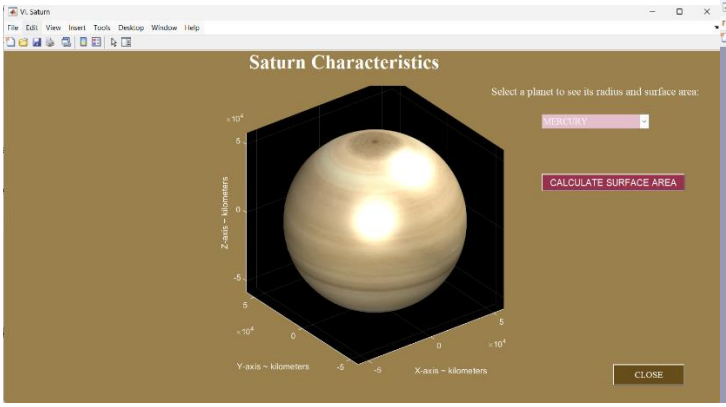
The rest of the windows:

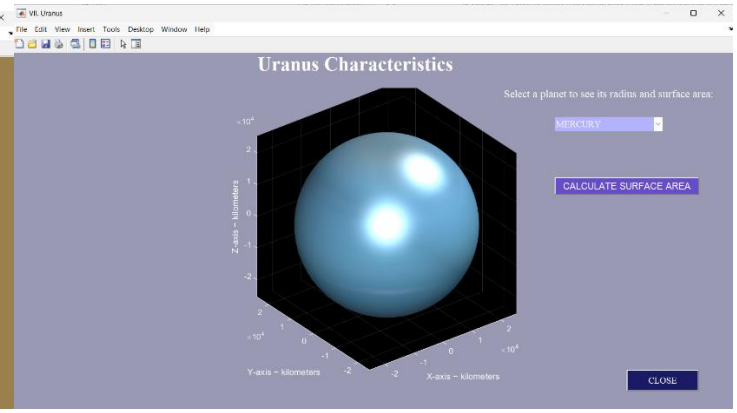


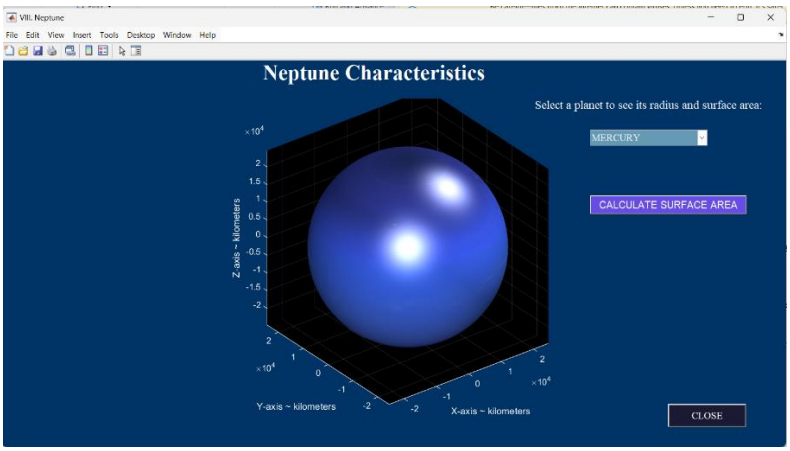




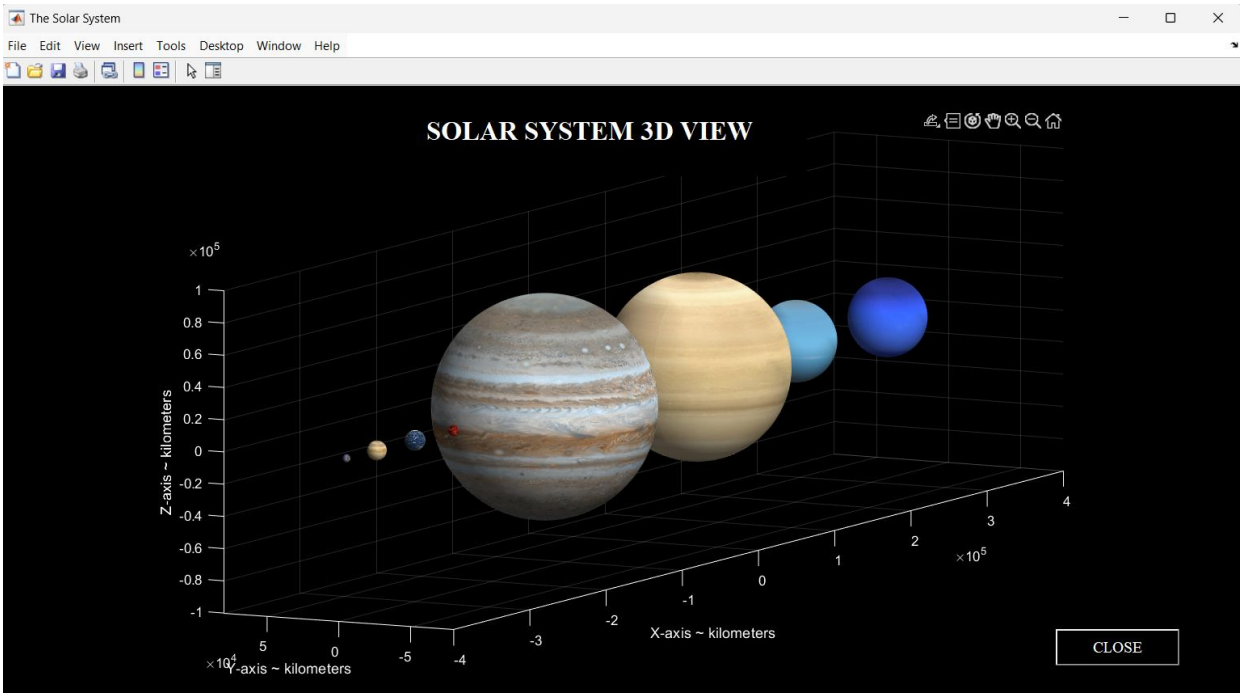








-The **3D view** of the solar system (real- life scaled) allows us to rotate, minimize and maximize the graphic as we please



IV. Files in Matlab

○ Main file

```
%%
% |%%%% MAIN MENU|
close all;

%creating the main menu
Fig=figure('Name','CAG Project ~ Negrea Sara',...
    'Units','normalized',...
    'Position',[0.1 0.1 0.8 0.7],...
    'color','none',...
    'NumberTitle','off');
backgroundimg =imread('milkyway.jpg');   %imread() allows us to 'read' the image

backgroundAxes = axes('Parent', Fig, 'Position', [0, 0, 1, 1]);
%%
imagesc(backgroundimg);    %this one sets the previous image as a backround
set(backgroundAxes, 'handlevisibility', 'off', 'visible', 'off');

uicontrol('Style','text',...
    'Units','normalized',...
    'Position',[0.3 0.85 0.35 0.1],...
    'backgroundcolor','#FFFFFF',...',...
    'foregroundcolor',"#330066",...
    'Fontname','Times New Roman',...
    'FontWeight','Bold',...
    'FontSize',30,...
    'String','The Solar System');

uicontrol('Style', 'text', ...
    'Units', 'normalized', ...
    'Position', [0.45 0.75 0.2 0.05], ...
    'BackgroundColor', '#FFFFFF', ...
    'ForegroundColor', "#330066", ...
    'FontName', 'Times New Roman', ...
    'FontWeight', 'Bold', ...
    'FontSize', 14, ...
    'HorizontalAlignment', 'center', ...
    'String', 'by Negrea Sara Diana');


%%
```

```
% _%%mercury_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.8 0.16 0.06],...
    'string','I. MERCURY',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','mercury_interface');

%%
% _%venus_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.7 0.16 0.06],...
    'string','II. VENUS',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','venus_interface');

%%
% _%%earth_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.6 0.16 0.06],...
    'string','III. EARTH',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','earth_interface');

%%
% _%%mars_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.5 0.16 0.06],...
    'string','IV. MARS',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','mars_interface');

%%
```

```
% _%%jupiter_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.4 0.16 0.06],...
    'string','V. JUPITER',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','jupiter_interface');

%%
% _%saturn_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.3 0.16 0.06],...
    'string','VI. SATURN',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','saturn_interface');
%%
% _
%uranus_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.2 0.16 0.06],...
    'string','VII. URANUS',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','uranus_interface');

%%
% _%%neptune_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.05 0.1 0.16 0.06],...
    'string','VIII. NEPTUNE',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0.42, 0.35, 0.63],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','neptune_interface');
```

```
%%
% _%display_solar_system_button_
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.35 0.45 0.3 0.12],...
    'string','DISPLAY SOLAR SYSTEM',...
    'FontName','Times New Roman',...
    'FontSize',14,...
    'backgroundcolor',[0.9, 0.6, 0.6],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','solarsyst');

%%documentation
uicontrol('style','pushbutton',...
    'Units','normalized',...
    'Position',[0.85 0.15 0.10 0.06],...
    'string','~documentation~',...
    'FontName','Times New Roman',...
    'FontSize',12,...
    'backgroundcolor',[0, 0.0, 0.3],...',...
    'foregroundcolor',"#FFFFFF",...
    'Callback','open (''the_solar_system.docx'')');
```

- ## Mercury window

```
    close all;

    Fig=figure('Name','I. Mercury',...
               'Units','normalized',...
               'Position',[0.1 0.1 0.8 0.7],...
               'color',[0 0.0 0.3],...
               'NumberTitle','off');
    radius_mercury();
    %title
    uicontrol('Style','text',...
            'Units','normalized',...
            'Position',[0.3 0.9 0.35 0.1],...
            'backgroundcolor',[0 0.0 0.3],...',...
            'foregroundcolor','white',...
            'Fontname','Times New Roman',...
            'FontWeight','Bold',...
            'FontSize',25,...
            'String','Mercury Characteristics');
```

```
img = imread('mercury_colormap.jpg'); %texture for Mercury

%creating the sphere
[x, y, z] = sphere(100);

scale_x = 2440;
scale_y = 2440;
scale_z = 2440;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';

view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

Venus window

```
close all;

Fig=figure('Name','II. Venus',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0.8 0.7 0.4],...
           'NumberTitle','off');

radius_venus();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0.8 0.7 0.4],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Venus Characteristics');

img = imread('venus_colormap.jpg'); %texture for Venus

[x, y, z] = sphere(100);

scale_x = 6052;
scale_y = 6052;
scale_z = 6052;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

```
view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

- **Earth window**

```
close all;

Fig=figure('Name','III. Earth',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0 0.2 0],...
           'NumberTitle','off');

radius_earth();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0 0.2 0],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Earth Characteristics');

img = imread('earth_colormap.jpg'); %texture for Earth

[x, y, z] = sphere(100);
```

```
scale_x = 6371;
scale_y = 6371;
scale_z = 6371;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';

view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

Mars window

```
close all;

Fig=figure('Name','IV. Mars',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0.4 0 0],...
           'NumberTitle','off');

radius_mars();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0.4 0 0],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Mars Characteristics');

img = imread('mars_colormap.jpg'); %texture for Earth

[x, y, z] = sphere(100);

scale_x = 3390;
scale_y = 3390;
scale_z = 3390;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

```
view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

○ Jupiter window

```
close all;

Fig=figure('Name','V. Jupiter',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0.68, 0.85, 0.95],...
           'NumberTitle','off');

radius_jupiter();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0.68, 0.85, 0.95],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Jupiter Characteristics');

img = imread('jupiter_colormap.jpg'); %texture for Earth
```

```
[x, y, z] = sphere(100);

scale_x = 69911;
scale_y = 69911;
scale_z = 69911;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';

view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

Saturn window

```
close all;

Fig=figure('Name','VI. Saturn',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0.6 0.5 0.3],...
           'NumberTitle','off');

radius_saturn();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0.6 0.5 0.3],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Saturn Characteristics');

img = imread('saturn_colormap.jpg'); %texture for Earth

[x, y, z] = sphere(100);

scale_x = 58232;
scale_y = 58232;
scale_z = 58232;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

```
view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

○ Uranus window

```
close all;

Fig=figure('Name','VII. Uranus',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0.6, 0.6, 0.7],...
           'NumberTitle','off');

radius_uranus();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0.6, 0.6, 0.7],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Uranus Characteristics');

img = imread('uranus_colormap.jpg'); %texture for Earth
```

```
[x, y, z] = sphere(100);

scale_x = 25362;
scale_y = 25362;
scale_z = 25362;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';

view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

○ Neptune window

```
close all;

Fig=figure('Name','VIII. Neptune',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color',[0 0.2 0.4],...
           'NumberTitle','off');
radius_neptune();
uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.9 0.35 0.1],...
        'backgroundcolor',[0 0.2 0.4],...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',25,...
        'String','Neptune Characteristics');

img = imread('neptune_colormap.jpg'); %texture for Earth

[x, y, z] = sphere(100);

scale_x = 24622;
scale_y = 24622;
scale_z = 24622;

x = scale_x * x;
y = scale_y * y;
z = scale_z * z;


surface(x, y, z, ...
    'FaceColor', 'texturemap', ...
    'EdgeColor', 'none', ...
    'CData', img);

axis equal;
ax = gca;

ax.Color = 'black';
ax.XColor = 'white';
ax.YColor = 'white';
```

```
ax.ZColor = 'white';

view(3);
xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
camlight headlight;


xlim([-scale_x, scale_x]);
ylim([-scale_y, scale_y]);
zlim([-scale_z, scale_z]);
ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';
```

- **Functions for determining the radius and the surface:**

- **Mercury**

```
function radius_mercury()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.6, 0.85, 0.4, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0 0 0.3], ...
              'FontSize', 16, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');
```

```
    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [0.4 0 1], ...
                      'ForegroundColor', 'white', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0 0 0.3], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                           'Units', 'normalized', ...
                           'Position', [0.75, 0.6, 0.2, 0.05], ...
                           'FontSize', 12, ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.7 0.6 0], ...
                           'String', 'CALCULATE SURFACE AREA', ...
                           'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.45, 0.3, 0.05], ...
                           'BackgroundColor', [0 0 0.3], ...
                           'ForegroundColor', 'white', ...
                           'FontName', 'Times New Roman', ...
                           'FontWeight', 'Bold', ...
                           'FontSize', 15, ...
                           'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0 0.0 0.2],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
```

```
                                    planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- ## Venus

```
function radius_venus()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0.8 0.7 0.4], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [0.5 0.4 0.2], ...
                      'ForegroundColor', 'white', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.8 0.7 0.4], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
```

```
                            'Units', 'normalized', ...
                            'Position', [0.75, 0.6, 0.2, 0.05], ...
                            'FontSize', 12, ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [0.7 0.6 0], ...
                            'String', 'CALCULATE SURFACE AREA', ...
                            'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.45, 0.3, 0.05], ...
                            'BackgroundColor', [0.8 0.7 0.4], ...
                            'ForegroundColor', 'white', ...
                            'FontName', 'Times New Roman', ...
                            'FontWeight', 'Bold', ...
                            'FontSize', 15, ...
                            'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.3 0.2 0],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

Earth

```
function radius_earth()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0 0.2 0], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [0 0 0.4], ...
                      'ForegroundColor', 'white', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0 0.2 0], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                           'Units', 'normalized', ...
                           'Position', [0.75, 0.6, 0.2, 0.05], ...
                           'FontSize', 12, ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.2 0.5 0.7], ...
                           'String', 'CALCULATE SURFACE AREA', ...
                           'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.45, 0.3, 0.05], ...
                           'BackgroundColor', [0 0.2 0], ...
                           'ForegroundColor', 'white', ...
                           'FontName', 'Times New Roman', ...
```

```
                            'FontWeight', 'Bold', ...
                            'FontSize', 15, ...
                            'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.3 0.4 0],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- ## Mars

```
function radius_mars()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0.4 0 0], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
```

```
                        'Units', 'normalized', ...
                        'Position', [0.75, 0.74, 0.15, 0.08], ...
                        'String', planets, ...
                        'FontName', 'Times New Roman', ...
                        'FontSize', 12, ...
                        'HorizontalAlignment', 'center',...
                        'BackgroundColor', [0.9 0.2 0], ...
                        'ForegroundColor', 'white', ...
                        'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.68, 0.25, 0.05], ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [0.4 0 0], ...
                            'FontName', 'Times New Roman', ...
                            'FontSize', 14, ...
                            'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                            'Units', 'normalized', ...
                            'Position', [0.75, 0.6, 0.2, 0.05], ...
                            'FontSize', 12, ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [1 0.4 0.2], ...
                            'String', 'CALCULATE SURFACE AREA', ...
                            'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.45, 0.3, 0.05], ...
                            'BackgroundColor', [0.4 0 0], ...
                            'ForegroundColor', 'white', ...
                            'FontName', 'Times New Roman', ...
                            'FontWeight', 'Bold', ...
                            'FontSize', 15, ...
                            'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.2 0 0],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
```

```
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- Jupiter

```
function radius_jupiter()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0.68, 0.85, 0.95], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [1, 1, 0.9], ...
                      'ForegroundColor', 'black', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.68, 0.85, 0.95], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');
```

```
    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                           'Units', 'normalized', ...
                           'Position', [0.75, 0.6, 0.2, 0.05], ...
                           'FontSize', 12, ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.8, 0.6, 0.5], ...
                           'String', 'CALCULATE SURFACE AREA', ...
                           'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.45, 0.3, 0.05], ...
                           'BackgroundColor', [0.68, 0.85, 0.95], ...
                           'ForegroundColor', 'white', ...
                           'FontName', 'Times New Roman', ...
                           'FontWeight', 'Bold', ...
                           'FontSize', 15, ...
                           'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.3, 0.4, 0.8],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                     planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                     planets{selectedIndex}, surfaceArea);
    end
end
```

- Saturn

```
function radius_saturn()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0.6 0.5 0.3], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [0.9, 0.75, 0.8], ...
                      'ForegroundColor', 'white', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.6 0.5 0.3], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                           'Units', 'normalized', ...
                           'Position', [0.75, 0.6, 0.2, 0.05], ...
                           'FontSize', 12, ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0.6, 0.2, 0.3], ...
                           'String', 'CALCULATE SURFACE AREA', ...
                           'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.45, 0.3, 0.05], ...
                           'BackgroundColor', [0.6 0.5 0.3], ...
                           'ForegroundColor', 'white', ...
                           'FontName', 'Times New Roman', ...
                           'FontWeight', 'Bold', ...
                           'FontSize', 15, ...
                           'String', '');

    %closebutton
```

```
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.4 0.3 0.1],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- **Uranus**

```
function radius_uranus()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0.6, 0.6, 0.7], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
```

```
                        'Position', [0.75, 0.74, 0.15, 0.08], ...
                        'String', planets, ...
                        'FontName', 'Times New Roman', ...
                        'FontSize', 12, ...
                        'HorizontalAlignment', 'center',...
                        'BackgroundColor', [0.7, 0.7, 1], ...
                        'ForegroundColor', 'white', ...
                        'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.68, 0.25, 0.05], ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [0.6, 0.6, 0.7], ...
                            'FontName', 'Times New Roman', ...
                            'FontSize', 14, ...
                            'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                            'Units', 'normalized', ...
                            'Position', [0.75, 0.6, 0.2, 0.05], ...
                            'FontSize', 12, ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [0.4 0.3 0.8], ...
                            'String', 'CALCULATE SURFACE AREA', ...
                            'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.45, 0.3, 0.05], ...
                            'BackgroundColor', [0.6, 0.6, 0.7], ...
                            'ForegroundColor', 'white', ...
                            'FontName', 'Times New Roman', ...
                            'FontWeight', 'Bold', ...
                            'FontSize', 15, ...
                            'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.1 0.1 0.4],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
    end
```

```
    function calculate_surface_area(~, ~)
        selectedIndex = popup.Value;
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- Neptune

```
function radius_neptune()

    planets = {'MERCURY', 'VENUS', 'EARTH', 'MARS', 'JUPITER', 'SATURN', 'URANUS', 'NEPTUNE'};
    radii = [2439, 6051.8, 6371, 3389, 69911, 58232, 25362, 24622]; % Radii in km

    uicontrol('Style', 'text', ...
              'Units', 'normalized', ...
              'Position', [0.65, 0.85, 0.35, 0.05], ...
              'String', 'Select a planet to see its radius and surface area:', ...
              'ForegroundColor', 'white', ...
              'BackgroundColor', [0 0.2 0.4], ...
              'FontSize', 14, ...
              'FontName', 'Times New Roman', ...
              'HorizontalAlignment', 'center');

    popup = uicontrol('Style', 'popupmenu', ...
                      'Units', 'normalized', ...
                      'Position', [0.75, 0.74, 0.15, 0.08], ...
                      'String', planets, ...
                      'FontName', 'Times New Roman', ...
                      'FontSize', 12, ...
                      'HorizontalAlignment', 'center',...
                      'BackgroundColor', [0.4 0.6 0.7], ...
                      'ForegroundColor', 'white', ...
                      'Callback', @update_radius);

    % Text to display the selected planet's radius
radiusText = uicontrol('Style', 'text', ...
                           'Units', 'normalized', ...
                           'Position', [0.7, 0.68, 0.25, 0.05], ...
                           'ForegroundColor', 'white', ...
                           'BackgroundColor', [0 0.2 0.4], ...
                           'FontName', 'Times New Roman', ...
                           'FontSize', 14, ...
                           'HorizontalAlignment', 'center');

    %button for area
calcButton = uicontrol('Style', 'pushbutton', ...
                           'Units', 'normalized', ...
                           'Position', [0.75, 0.6, 0.2, 0.05], ...
```

```
                            'FontSize', 12, ...
                            'ForegroundColor', 'white', ...
                            'BackgroundColor', [0.4 0.3 0.9], ...
                            'String', 'CALCULATE SURFACE AREA', ...
                            'Callback', @calculate_surface_area);

%calculated area
    resultText = uicontrol('Style', 'text', ...
                            'Units', 'normalized', ...
                            'Position', [0.7, 0.45, 0.3, 0.05], ...
                            'BackgroundColor', [0 0.2 0.4], ...
                            'ForegroundColor', 'white', ...
                            'FontName', 'Times New Roman', ...
                            'FontWeight', 'Bold', ...
                            'FontSize', 15, ...
                            'String', '');

    %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor',[0.1 0.1 0.2],...',...
        'foregroundcolor','white',...
          'Callback','interface');


    function update_radius(~, ~)
        selectedIndex = popup.Value;
        selectedRadius = radii(selectedIndex);
        radiusText.String = sprintf('Radius of %s: %.1f km', ...
                                    planets{selectedIndex}, selectedRadius);
    end

    function calculate_surface_area(~, ~)
        % Get the selected planet index
        selectedIndex = popup.Value;

        % Get the radius of the selected planet
        radius = radii(selectedIndex);

        % Calculate surface area
        surfaceArea = 4 * pi * radius^2;

        % Update the result text
        resultText.String = sprintf('%s is %.2f km^2', ...
                                    planets{selectedIndex}, surfaceArea);
    end
end
```

- **Solar System window**

```
close all;

Fig=figure('Name','The Solar System',...
           'Units','normalized',...
           'Position',[0.1 0.1 0.8 0.7],...
           'color','black',...
           'NumberTitle','off');

uicontrol('Style','text',...
        'Units','normalized',...
        'Position',[0.3 0.85 0.35 0.1],...
        'backgroundcolor','black',...',...
        'foregroundcolor','white',...
        'Fontname','Times New Roman',...
        'FontWeight','Bold',...
        'FontSize',20,...
        'String','SOLAR SYSTEM 3D VIEW');

[x, y, z] = sphere(100);

% Planet Data (Name, Radius in kilometers, Texture File, Offset along x-axis)
planets = {
    'Mercury', 2440, 'mercury_colormap.jpg', -390000;
    'Venus', 6052, 'venus_colormap.jpg', -350000;
    'Earth', 6371, 'earth_colormap.jpg', -300000;
    'Mars', 3390, 'mars_colormap.jpg', -250000;
    'Jupiter', 69911, 'jupiter_colormap.jpg', -130000;
    'Saturn', 58232, 'saturn_colormap.jpg', 70000;
    'Uranus', 25362, 'uranus_colormap.jpg', 200000;
    'Neptune', 24622, 'neptune_colormap.jpg', 320000;
};

for i = 1:size(planets, 1)
    % Extract planet data
    planet_name = planets{i, 1};
    radius = planets{i, 2};
    texture_file = planets{i, 3};
    x_offset = planets{i, 4};

    img = imread(texture_file);

    x_scaled = radius * x + x_offset; % Apply x-axis offset
    y_scaled = radius * y;
    z_scaled = radius * z;

    surface(x_scaled, y_scaled, z_scaled, ...
        'FaceColor', 'texturemap', ...
        'EdgeColor', 'none', ...
        'CData', img);
end

%axis properties
axis equal;
ax = gca; % Get current axes handle
ax.Color = 'black'; % Set axes background to black
ax.XColor = 'white'; % Set x-axis color to white
```

```
ax.YColor = 'white'; % Set y-axis color to white
ax.ZColor = 'white'; % Set z-axis color to white

view(3);

xlabel('X-axis ~ kilometers', 'Color', 'white');
ylabel('Y-axis ~ kilometers', 'Color', 'white');
zlabel('Z-axis ~ kilometers', 'Color', 'white');

grid on;
ax.GridColor = 'white';

light;
lighting phong;
material dull;
camlight headlight;

xlim([-400000, 400000]);
ylim([-80000, 80000]);
zlim([-100000, 100000]);

ax.TickLabelInterpreter = 'none';
ax.XColor = 'white';
ax.YColor = 'white';
ax.ZColor = 'white';

 %closebutton
 uicontrol('style','pushbutton',...
          'Units','normalized',...
          'Position',[0.85 0.05 0.10 0.06],...
          'string','CLOSE',...
          'FontName','Times New Roman',...
          'FontSize',12,...
           'backgroundcolor','black',...',...
        'foregroundcolor','white',...
          'Callback','interface');
```

Bibliography:

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